

ADAPTING SURVEY DESIGNS FOR VECTOR SURVEILLANCE USING BAYESIAN DECISION THEORY

AN APPLICATION TO AN ONGOING SOUTH CAROLINA TICK MONITORING PROGRAM

Bren Case

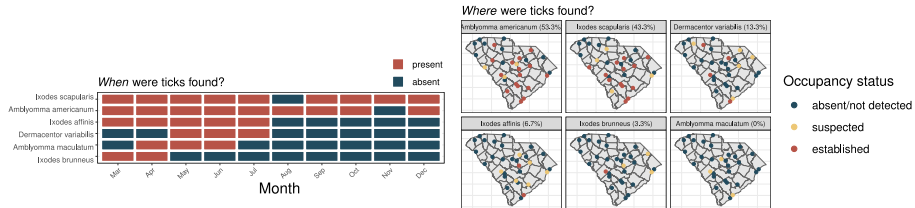
10 February 2023

Department of Computer Science
University of Vermont
brendandaisy.github.io



Tick surveillance in South Carolina

- Limited historical data for southeastern US
 - ▶ Low disease burden leads to lack of motivation and funding ¹
 - ▶ **Proactive surveillance** nonetheless an important part of informing trends and preparing for future ²

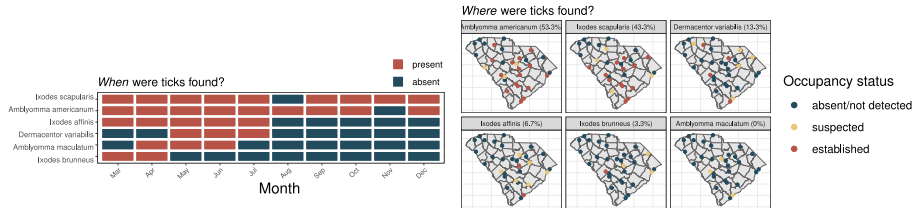


1. DYE-BRAUMULLER et al., "Needs Assessment of Southeastern United States Vector Control Agencies", *Tropical Medicine and Infectious Disease*. 2022.

2. EISEN et PADDOCK, "Tick and Tickborne Pathogen Surveillance as a Public Health Tool in the United States", *Journal of Medical Entomology*. 2021.

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 - ▶ Low disease burden leads to lack of motivation and funding ¹
 - ▶ **Proactive surveillance** nonetheless an important part of informing trends and preparing for future ²
- Joint effort between UofSC and SC Department of Health (DHEC)
 - ▶ Includes state parks, animal shelters, citizen science
 - ▶ Expanding with each year and funding for next 5



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- Previous tick distribution maps had highest uncertainty in southeastern US^{3, 4}

3. HAHN et al., "Modeling the Geographic Distribution of Ixodes Scapularis and Ixodes Pacificus (Acari)", *Journal of Medical Entomology*. 2016.

4. SPRINGER et al., "Modeling the Present and Future Geographic Distribution of the Lone Star Tick, Amblyomma Americanum (Ixodida)", *The American Journal of Tropical Medicine and Hygiene*. 2015.

5. VECTOR-BORNE DISEASES, *Guide to the Surveillance of Metastriate Ticks (Acari : Ixodidae) and their Pathogens in the United States*, 2020.

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- Previous tick distribution maps had highest uncertainty in southeastern US^{3, 4}
- Sampling public natural areas is good proxy for exposure, but expensive and error-prone⁵
- High variability inherent in tick collection
 - ▶ Some species hard to find depending on questing behaviour and collection method
 - ▶ Complex population dynamics lead to variable outcomes

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How can we establish a surveillance program that eliminates as much of this uncertainty as possible?

- Decision theoretic criteria to propose future locations/times for surveillance
- Park collections data from 2020-2021 to inform future outcomes
- Latest computational tools to bring Bayesian Experimental Design to Big Data setting

1. Specify and fit a model to assess parameter uncertainty from initial data $\mathbf{y}_{\text{init}}$
2. For candidate design $\mathbf{d}$, assign **utility** $U(\mathbf{d})$:
 - ▶ Utility $U(\mathbf{d}, \mathbf{y})$ for each potential outcome resulting from $\mathbf{d}$
 - ▶ Expected utility over possible outcomes : ⁶

$$U(\mathbf{d}) = \int U(\mathbf{d}, \mathbf{y}) P(\mathbf{y} \mid \mathbf{d}, \mathbf{y}_{\text{init}}) d\mathbf{y}$$

3. Optimize $U(\mathbf{d})$:

$$\mathbf{d}^* = \operatorname{argmax}_{\mathbf{d}} U(\mathbf{d})$$

6. DORAZIO et JOHNSON, "Bayesian Inference and Decision Theory—a Framework for Decision Making in Natural Resource Management", *Ecological Applications*. 2003.

Risk during visit at site i , month t , and species j :

The diagram illustrates the components of a regression-based species distribution model. It shows three main components being summed together to calculate the logit of the risk r_{ijt} .

Component 1: Environmental effects

- Input: Environmental variables (monthly temperature, tree cover, ...)
- Effect: Environmental effects
- Formula: $\sum_p f_{ij}(x_{pt})$
- Description: (nonlinear) correlations with environment

Component 2: Spatiotemporal effects

- Input: Survey variables (month, date)
- Effect: Spatiotemporal effects
- Formula: $s_{ij} + m_{jt}$
- Description: correlations over space/time

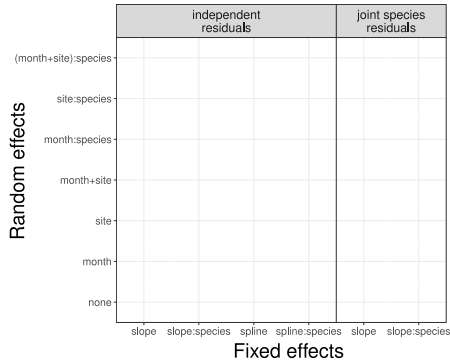
Component 3: Joint residual effects

- Input: Species ID
- Effect: Joint residual effects
- Formula: e_{ijt}
- Description: correlation between species

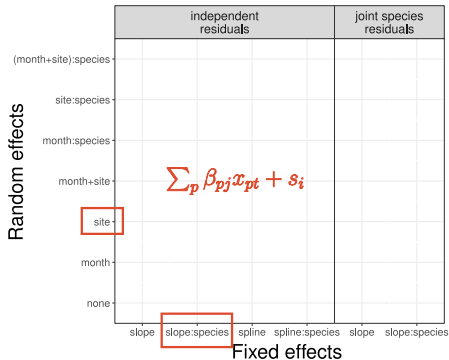
The overall equation is:

$$\text{logit}(r_{ijt}) = \text{Environmental effects} + \text{Spatiotemporal effects} + \text{Joint residual effects}$$

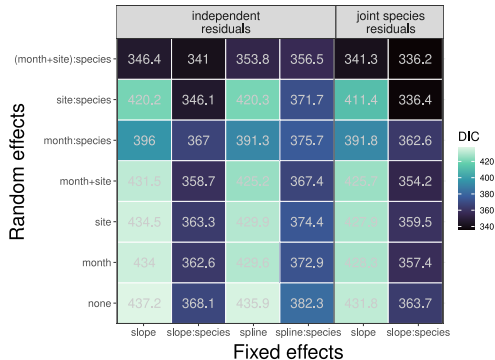
- Models compared using Deviance Information Criterion (DIC)



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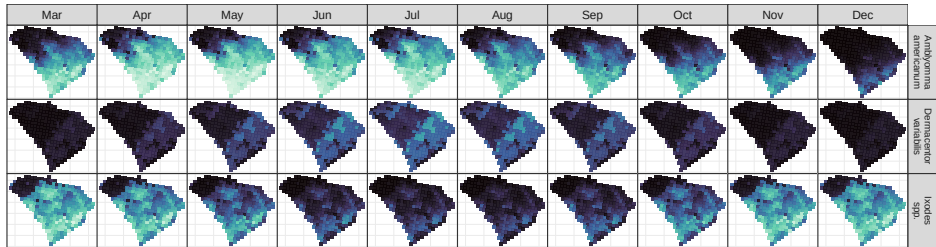


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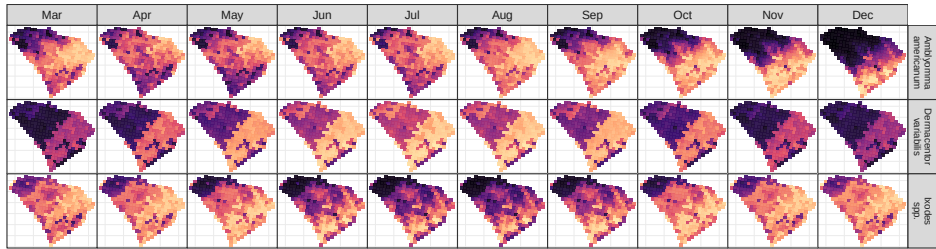
- ▷ Species-dependent effects clearly better
- ▷ Limited geographical diversity could explain poor performance of spline models

Predictions given 2020 and 2021 data



Probability tick present

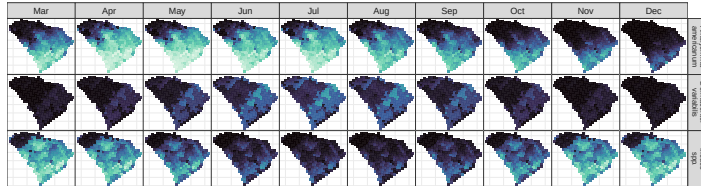
0.25 0.50 0.75



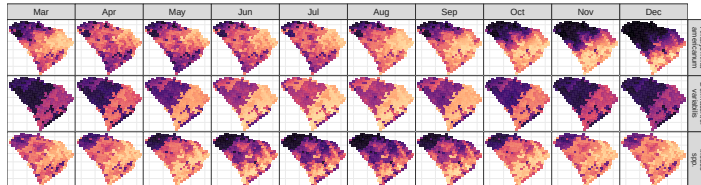
Standard deviation

0.0 0.1 0.2 0.3 0.4

Predictions given 2020 and 2021 data



Probability tick present
0.25 0.50 0.75



Standard deviation
0.0 0.1 0.2 0.3 0.4

▷ High predicted risk of *Amblyomma* and *Ixodes* in southeast, but different seasons

▷ Prediction uncertainty extremely high, especially for *Ixodes*

Recall the expected utility :

$$U(\mathbf{d}) = \int U(\mathbf{d}, \mathbf{y}) P(\mathbf{y} \mid \mathbf{d}, \mathbf{y}_0) d\mathbf{y}$$

7. KLEINEGESSE, DROVANDI et GUTMANN, "Sequential Bayesian Experimental Design for Implicit Models via Mutual Information", *arXiv :2003.09379 [cs, stat]*. 2020.

8. CHALONER et VERDINELLI, "Bayesian Experimental Design : A Review", *Institute of Mathematical Statistics*. 1995.

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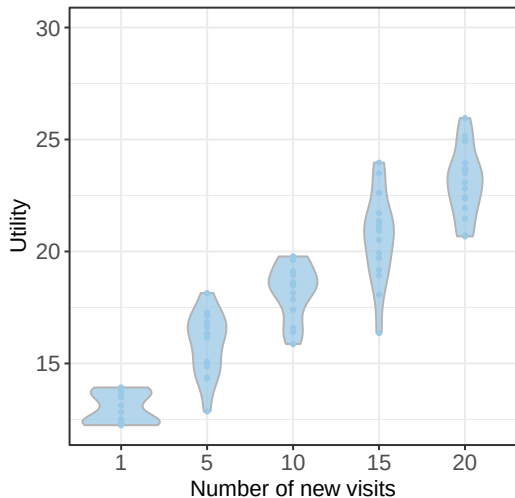
$$U(\mathbf{d}) = \int U(\mathbf{d}, \mathbf{y}) P(\mathbf{y} \mid \mathbf{d}, \mathbf{y}_0) d\mathbf{y}$$

Bayesian D-optimality⁷ for non-reducible uncertainty :⁸

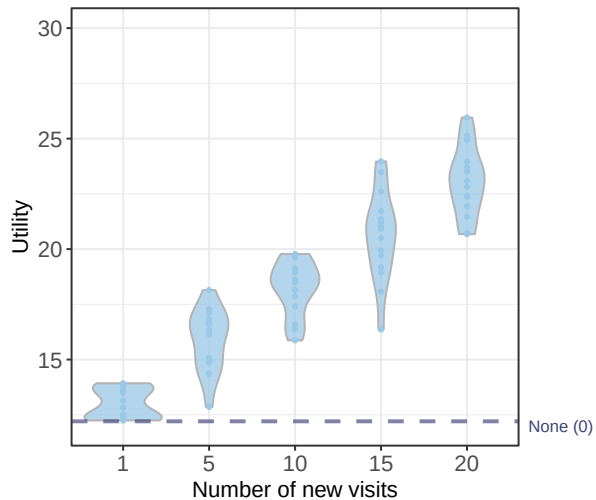
$$U(\mathbf{d}, \mathbf{y}) = -\log(\det(\boldsymbol{\beta} \mid \mathbf{y}_0, \mathbf{y}))$$

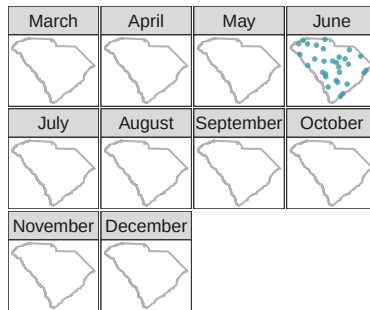
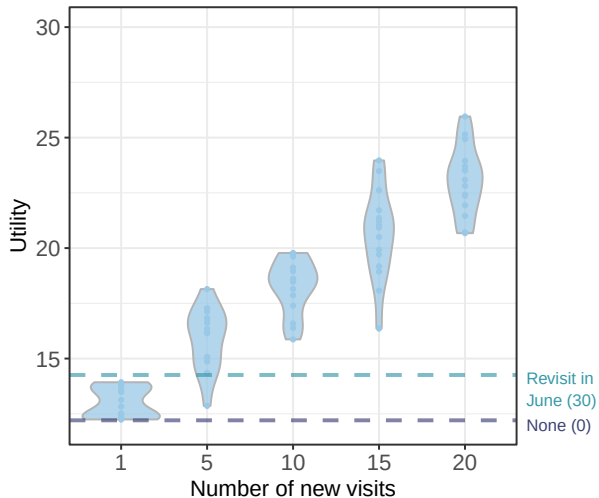
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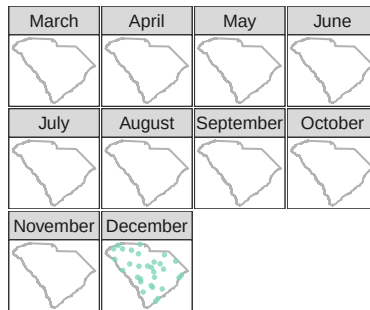
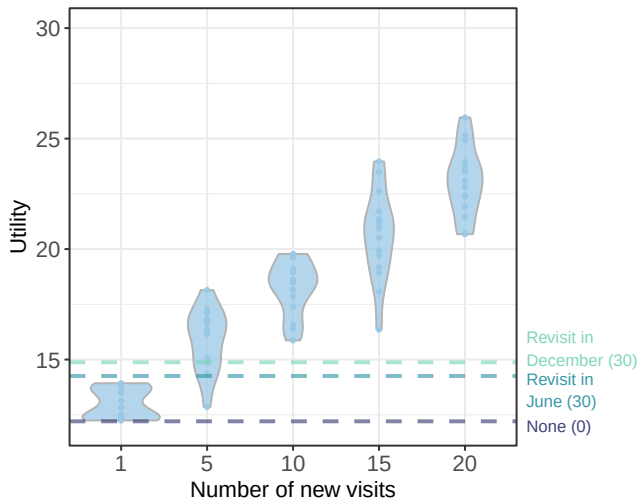
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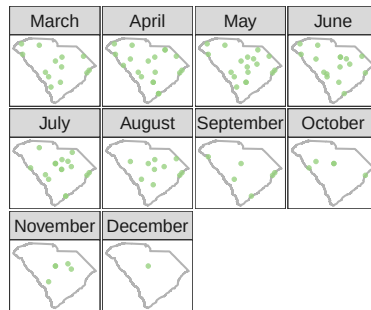
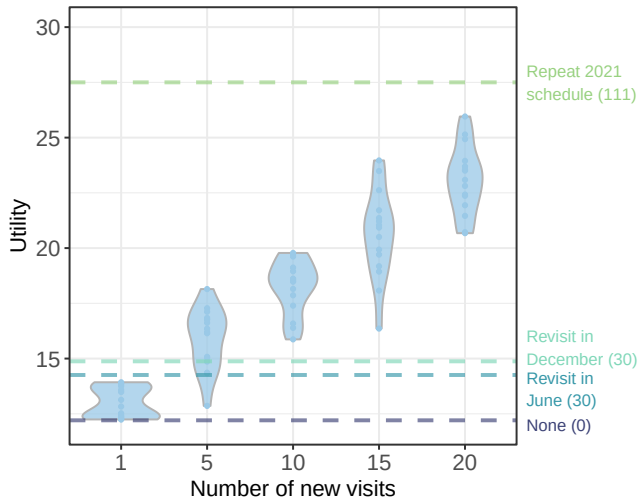
▷ $U(\mathbf{d})$ approximated for a random selection of 20 collection schedules, for increasing number of new visits

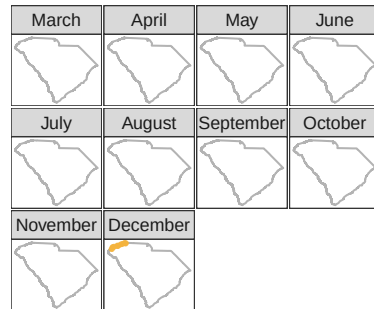
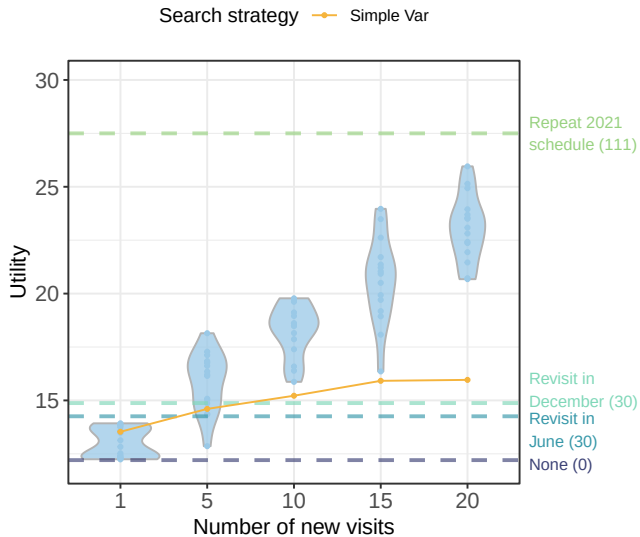




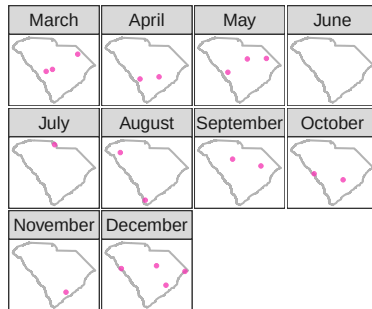
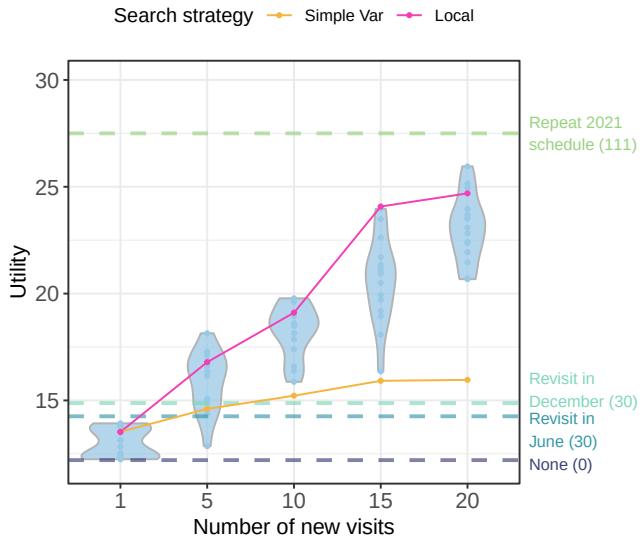


Results



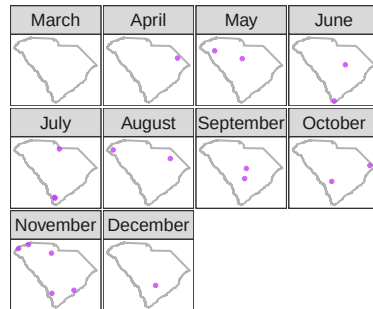
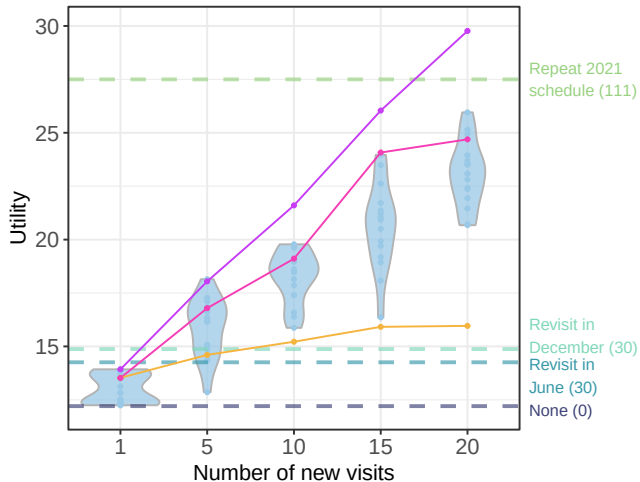


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Search strategy — Simple Var — Local — Simulated Annealing



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 - ▶ Municipal parks vs. campgrounds
- Weight locations based on human traffic
 - ▶ Incorporate GPS data?

Collaborators

Kyndall C. Dye-Braumuller, Chris Evans, Huixuan Li, Lauren Rustin, Melissa S. Nolan

Funding and computational resources

